TEXTBOOK EVALUATION

I. Choose the correct answer

1. According to Mendel alleles have the following character
   a) Pair of genes
   b) Responsible for character
   c) Production of gametes
   d) Recessive factors

2. 9 : 3 : 3 : 1 ratio is due to
   a) Segregation b) Crossing over
   c) Independent assortment d) Recessiveness

3. The region of the chromosome where the spindle fibres get attached during cell division
   a) Chromomere b) Centrosome
   c) Centromere d) Chromonema

4. The centromere is found at the centre of the _____ chromosome.
   a) Telocentric b) Metacentric
   c) Sub-metacentric d) Acrocentric

5. The _____ units form the backbone of the DNA.
   a) 5 carbon sugar b) Phosphate
   c) Nitrogenous bases d) Sugar phosphate

6. Okasaki fragments are joined together by _____.
   a) Helicase b) DNA polymerase
   c) RNA primer d) DNA ligase

7. The number of chromosomes found in human beings are _____.
   a) 22 pairs of autosomes and 1 pair of allosomes.
   b) 22 autosomes and 1 allosome
   c) 46 autosomes
   d) 46 pairs autosomes and 1 pair of allosomes.

8. The loss of one or more chromosome in a ploidy is called _____.
   a) Tetraploidy b) Aneuploidy
   c) Euploidy d) polyploidy

II. Fill in the blanks

1. The pairs of contrasting character (traits) of Mendel are called _____.
2. Physical expression of a gene is called _____
3. The thin thread like structures found in the nucleus of each cell are called _____.
4. DNA consists of two _____ chains
5. An inheritable change in the amount or the structure of a gene or a chromosome is called _____.

III. Identify whether the statement are True or False. Correct the false statement

1. A typical Mendelian dihybrid ratio of F_2 generation is 3:1.
2. A recessive factor is altered by the presence of a dominant factor.
3. Each gamete has only one allele of a gene.
4. Hybrid is an offspring from a cross between genetically different parent.
5. Some of the chromosomes have an elongated knob-like appendages known as telomere.
6. New nucleotides are added and new complementary strand of DNA is formed with the help of enzyme DNA polymerase.
7. Down's syndrome is the genetic condition with 45 chromosomes.

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